

Some interesting projects appear in these results, and it might be worth looking at a few. But don't spend too much time here, because we're about to create a visualization that will make the results much easier to read.

Monitoring API Rate Limits

Most APIs have *rate limits*, which means there's a limit to how many requests you can make in a certain amount of time. To see if you're approaching GitHub's limits, enter https://api.github.com/rate_limit into a web browser. You should see a response that begins like this:

```
{
  "resources": {
    --snip--
    ❶ "search": {
    ❷   "limit": 10,
    ❸   "remaining": 9,
    ❹   "reset": 1652338832,
      "used": 1,
      "resource": "search"
    },
    --snip--
```

The information we're interested in is the rate limit for the search API ❶. We see that the limit is 10 requests per minute ❷ and that we have 9 requests remaining for the current minute ❸. The value associated with the key "reset" represents the time in *Unix* or *epoch time* (the number of seconds since midnight on January 1, 1970) when our quota will reset ❹. If you reach your quota, you'll get a short response that lets you know you've reached the API limit. If you reach the limit, just wait until your quota resets.

NOTE

Many APIs require you to register and obtain an API key or access token to make API calls. As of this writing, GitHub has no such requirement, but if you obtain an access token, your limits will be much higher.

Visualizing Repositories Using Plotly

Let's make a visualization using the data we've gathered to show the relative popularity of Python projects on GitHub. We'll make an interactive bar chart: the height of each bar will represent the number of stars the project has acquired, and you'll be able to click the bar's label to go to that project's home on GitHub.